

INRIA FLOWERS / AIxIA

adtool

Assisted and Automated Discovery

A research workflow for AIxIA

Arthur Macdonald



Explore systems through their behaviors

A common research question

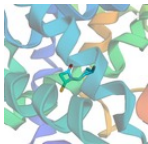
Which inputs produce different, interesting behaviors?



Particle Lenia

Rules and initial states

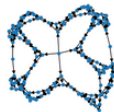
Patterns and motion



Molecular docking

Molecular candidates

Interaction patterns



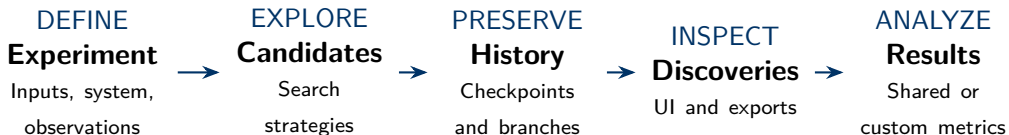
Hypergraph rewriting

Rewriting rules

Evolving structures

What adtool provides

A researcher-defined experiment moves through shared services

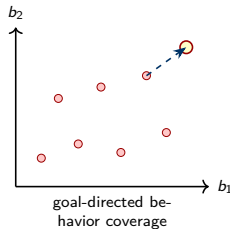
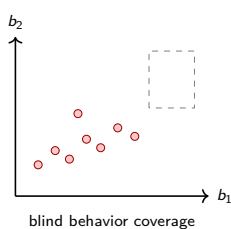


The experiment supplies domain knowledge; adtool carries the exploration, history, inspection, and analysis workflow.

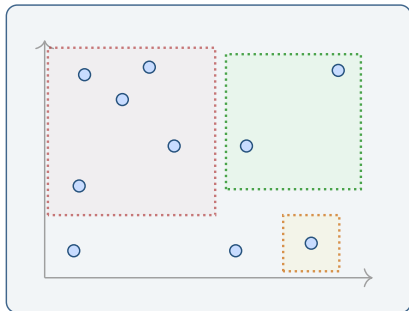
Why explore behavior space?

Intrinsically Motivated Goal Exploration Process

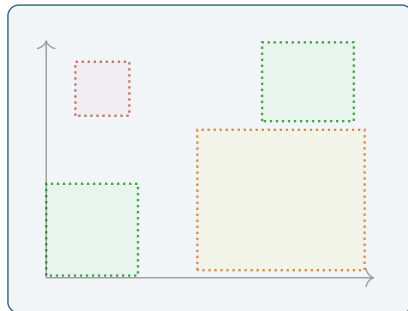
- Random inputs may reach only a clustered set of behaviors
- IMGEP selects goals in the observed behavior space
- Past experiments connect a behavior back to usable inputs



Parameter space



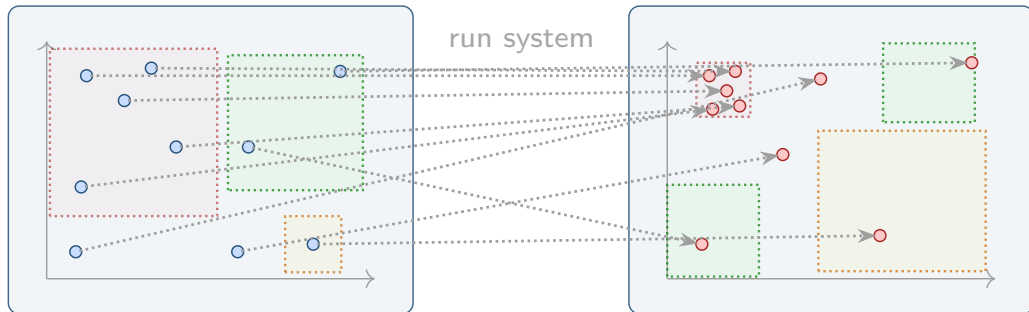
Behavior space



1. Sample parameters uniformly. The large red parameter region receives more random samples than the tiny yellow one.

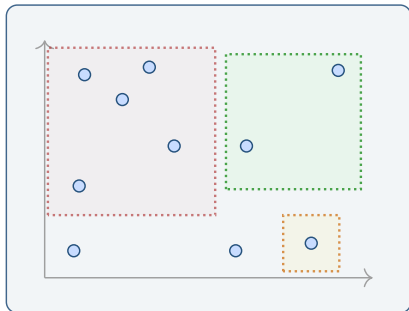
Parameter space

Behavior space

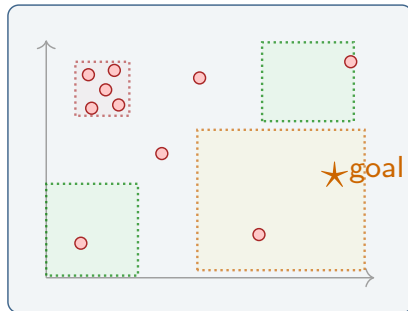


2. Run and describe. Red contracts and yellow expands: random parameter coverage becomes uneven behavior coverage.

Parameter space

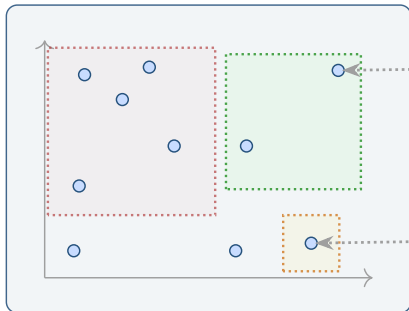


Behavior space

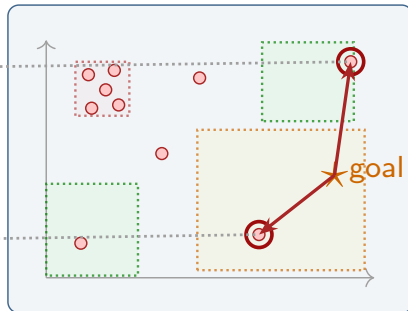


3. Sample a behavior goal. IMGEP can target an under-covered part of the large yellow behavior region.

Parameter space

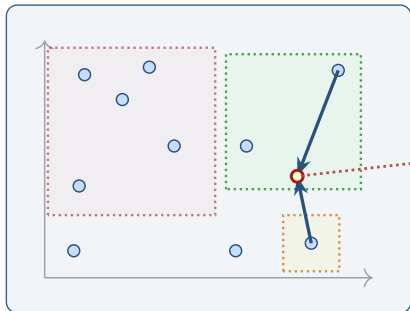


Behavior space

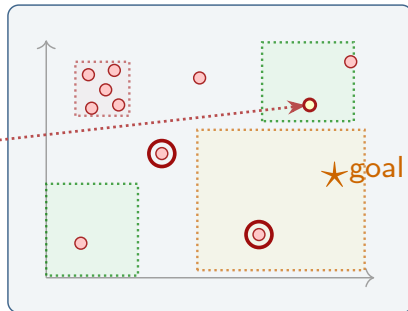


4. Retrieve. Find the closest archived behaviors and recover their associated parameters.

Parameter space

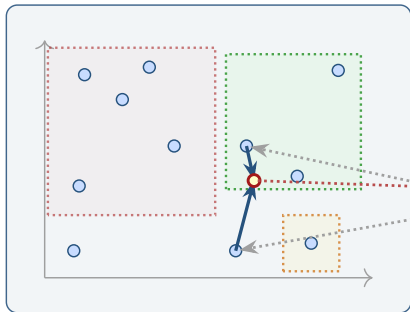


Behavior space

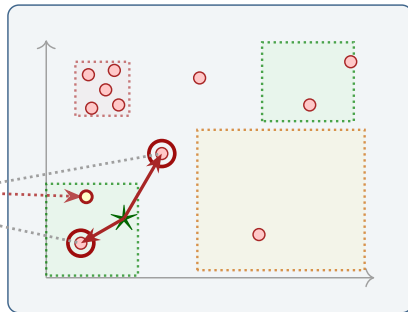


5. Mix and mutate. The candidate approaches the yellow goal, but does not reach it exactly. A missed goal is still information.

Parameter space



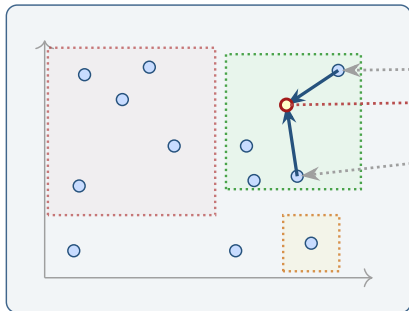
Behavior space



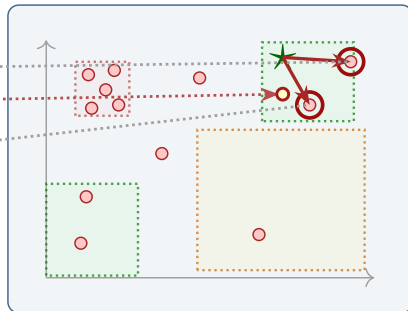
6. Iterate. Repeat the process with a new candidates.

IMGEP operational loop

Parameter space

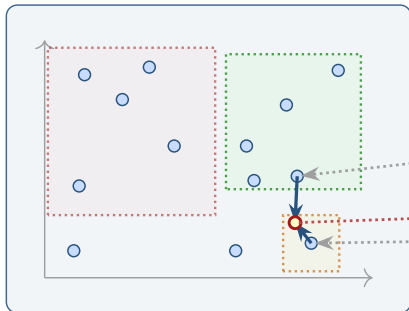


Behavior space

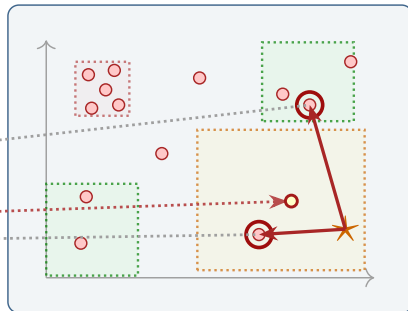


6. Iterate. Repeat the process with a new candidates.

Parameter space

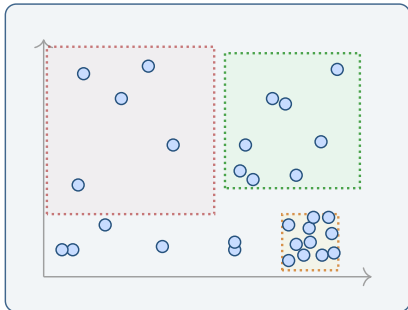


Behavior space

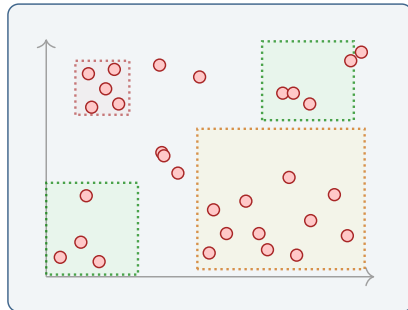


6. Iterate. Repeat the process with a new candidates.

Parameter space



Behavior space

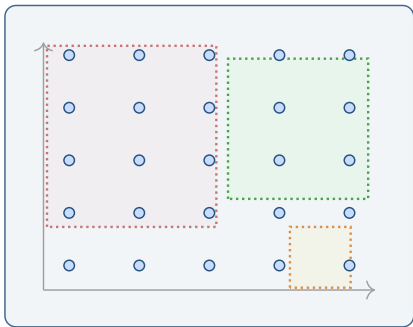


Result: The run allocates more trials to the small yellow parameter region, yielding the same emphasis in behavior space, while retaining discoveries outside the mapped zones.

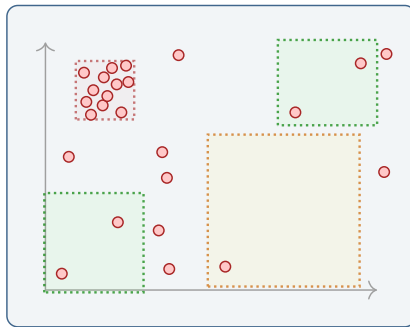
Same budget, uniformly covered parameter space

A regular 25-sample parameter grid is still uneven after the system mapping.

Parameter space



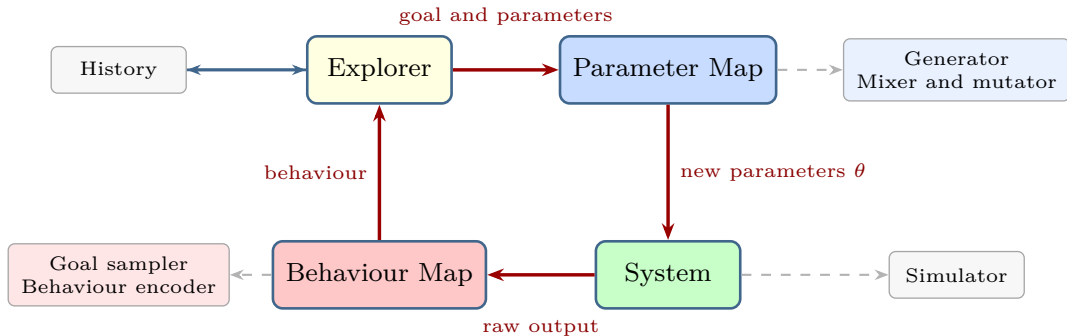
Behavior space



Every parameter-space cell receives one sample. In behavior space, many results still accumulate in the compact red region, while other regions receive few samples.

adtool: the IMGEP workflow

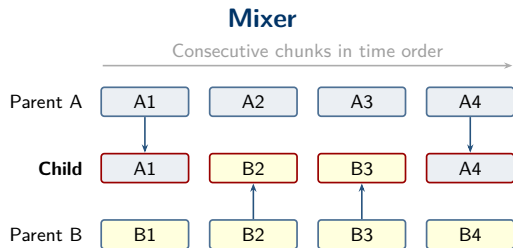
The algorithm mapped to reusable experiment components



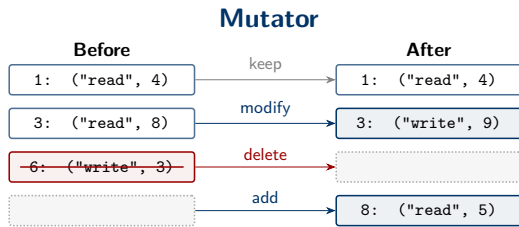
1. Choose a behavior goal
2. Vary a relevant past input
3. Run, describe, and keep it

Change one part of an experiment

The same experiment can use different ways to create or modify programs.



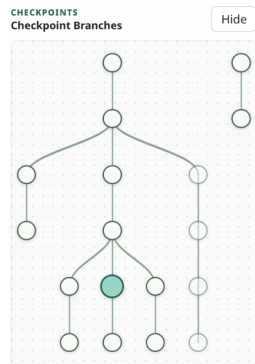
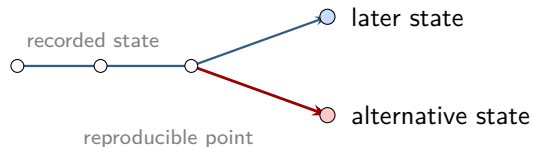
Combines chunks from two known candidate programs.



Retains, modifies, removes, or adds instructions.

Preserve experiment state

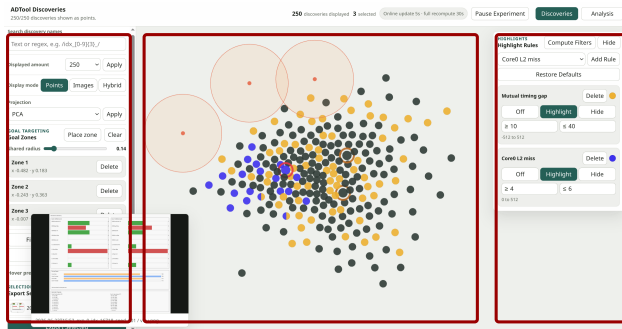
- Record a reproducible point in an exploration
- Return to that state without rebuilding artifacts
- Retain the provenance of later alternatives



A saved state in the current UI.

Inspect discoveries in the UI

The discovery workspace separates display controls, direct interaction, and filters.



Display
Set projection and
parameters

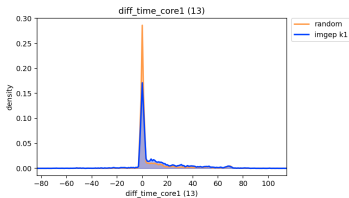
Discoveries
Navigate and inspect

Filters
Filter by metric or origin

Analyze discoveries in three complementary views

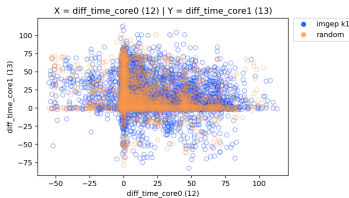
Default metrics available for analysis of saved discoveries.

1D comparison



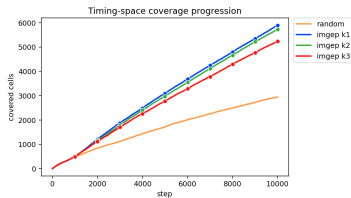
Which values are reached?

2D comparison



Which combinations appear?

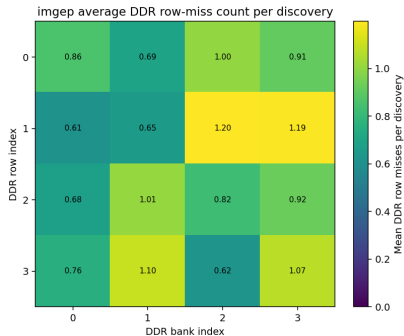
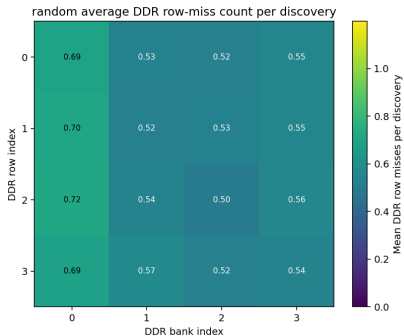
Coverage progression



How does explored diversity grow?

Add analysis suited to your scientific question

Metric to measure where do DDR row misses occur in the interference study.

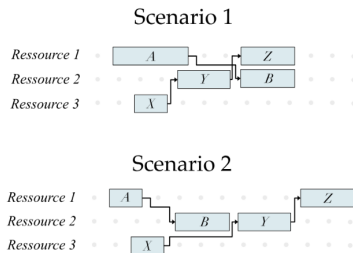


Artificial Intelligence for Interference Analysis

The identification question

Which memory-access patterns lead to different timing effects?

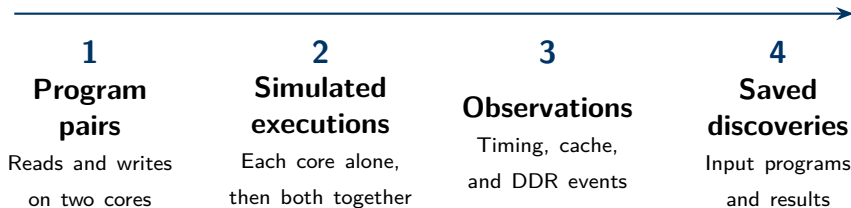
- Cores compete for shared resources
- Interactions change execution timing
- Diverse cases help researchers investigate



Timing illustration: C. Courtaud

The interference study inside adtool

From program pairs to saved discoveries



Specific to the study

Program rules, simulator, and observations.

Reusable through adtool

Program-based exploration support, checkpoints, UI, and analysis.

Why use adtool for AIxIA experiments?

Ad hoc experiment work

Run-specific scripts

Files and static plots

Repeat setup

With adtool

Configurable experiment components

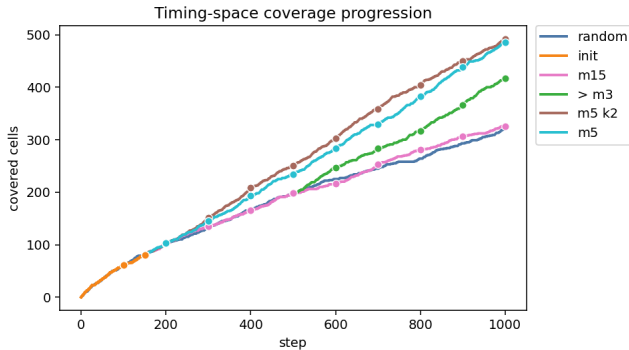
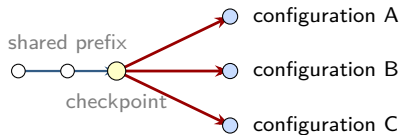
Discoveries to inspect and export

Branch a saved run

The AIxIA simulator remains study-specific.
adtool provides the reusable exploration, history, inspection, and comparison workflow.

AlxIA example: compare continuations from a checkpoint

One shared interference-exploration prefix becomes several directly comparable continuations.



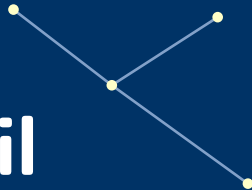
No need to repeat the shared initialization for every alternative.

Can manually adapt the experiment configuration mid-run based on current progress.

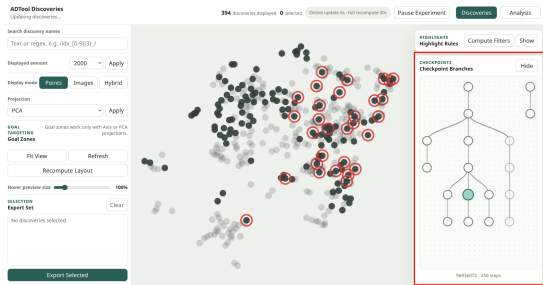
What AIxIA researchers can do with adtool



Further detail

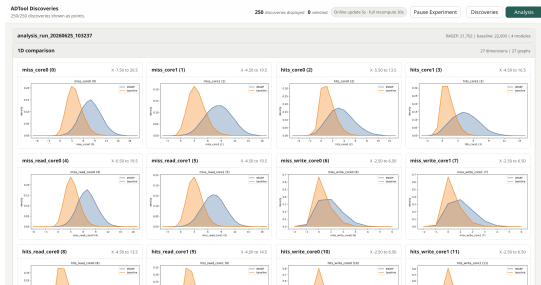


Checkpoint screen



Navigate an experiment's saved branches.

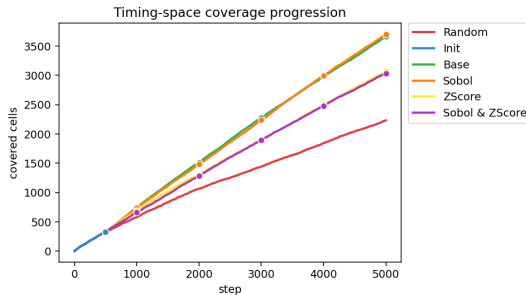
Analysis screen



Explore distributions and comparisons from saved discoveries.

Alternative components: a preliminary comparison

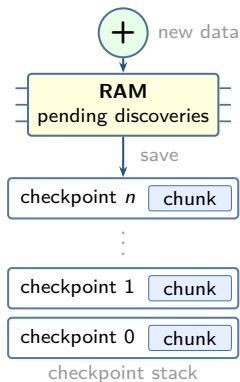
Evaluation of custom Sobol goal sampler and z-score encoder under the same exploration budget.



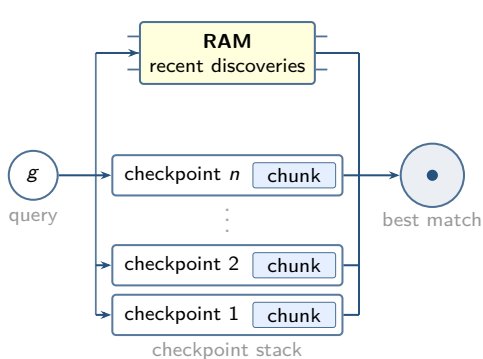
History implementation: RAM and persistent chunks

Recent discoveries stay in RAM while saved chunks make earlier discoveries available for retrieval.

Recording



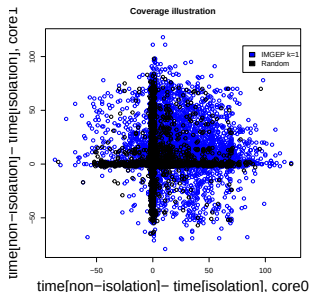
Retrieval



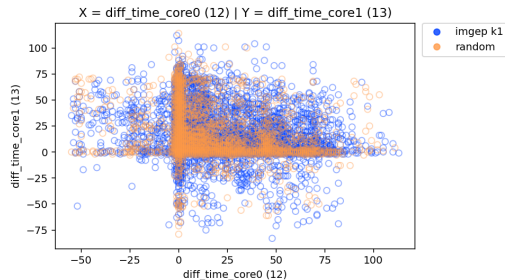
Validation against the reference study

The same analyses compare Ludovic Matar's source experiment workflow with the new adtool workflow.

Reference implementation



adtool integration



Independent stochastic runs

References and credits

Software, scientific background, and visual credits

adtool library. Inria FLOWERS, *adtool: Assisted and Automated Discovery for Complex Systems*.
github.com/flowersteam/adtool

IMGEP. Forestier, S., Portelas, R., Mollard, Y., & Oudeyer, P.-Y. (2022). *Intrinsically Motivated Goal Exploration Processes with Automatic Curriculum Learning*. Inria FLOWERS Bordeaux.

Interference study. Matar, L., & Moulin-Frier, C. (2026). *Curiosity-driven automated discovery of interference patterns in multi-core architectures*. Inria FLOWERS Bordeaux.

Figures and images. All figures and images not individually referenced are by Arthur Macdonald.